

One Plot, Two Reporting Strategies

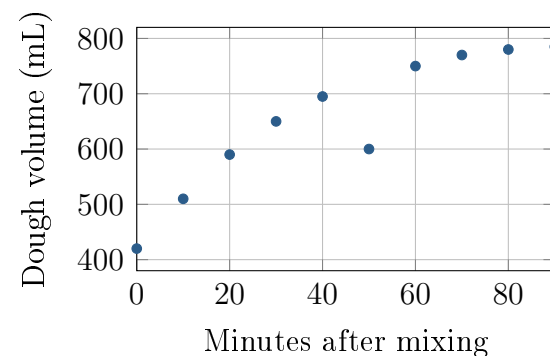
Unit 5 · Topic 5.1 · TODAY'S PROBLEM

Suppose a test kitchen measures dough volume for ten batches at assigned times after mixing. The scatterplot shows the paired measurements.

Jules: Use a straight-line summary because volume generally increases with time. Treat the 50-minute batch as ordinary variation.

Ravi: Decide the form from how the rate changes, and report the 50-minute batch separately if it departs from nearby values.

Selected measurements (minutes, mL): (0, 420), (20, 590), (70, 770), (90, 785).



FIRST TAKE (5 min, on your sheet)

Which reporting strategy seems more defensible, Jules's or Ravi's? Cite one graph feature.

- DOK 1** (a) Name the explanatory and response variables, with units.
- DOK 2** (b) Find the volume increase from 0 to 20 minutes and from 70 to 90 minutes. Is the 50-minute batch above or below the nearby batches?
- DOK 3** (c) Whose reporting strategy is supported? Describe the direction, form, strength, and unusual point in context, using your two increases. Explain how a straight-line summary could mislead someone predicting later growth. Write 3–4 sentences.

Video + follow-along: [u2_lesson4_live.html](https://www.khanacademy.org/a/u2_lesson4_live_html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

